TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	94.0 ± 0.63
Correlation Stability	89.0 ± 1.00
Distribution Consistency	96.6 ± 0.49
Entropy Preservation	96.2 ± 0.40
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	94.7 ± 1.49
Feature Accuracy	97.3 ± 0.64
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	3.922	+0.077	2.0%	OK
Glucose	120.895	122.223	+1.329	1.1%	OK
BloodPressure	69.105	69.899	+0.794	1.1%	OK
SkinThickness	20.536	20.886	+0.350	1.7%	OK
Insulin	79.799	80.953	+1.153	1.4%	OK
BMI	31.993	31.881	-0.111	0.3%	OK
DiabetesPedigreeFunction	0.472	0.494	+0.022	4.6%	OK
Age	33.241	33.459	+0.218	0.7%	OK
Outcome	0.349	0.353	+0.004	1.3%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	94.0 ± 0.63
Correlation Stability (*2)	89.0 ± 1.00
Distribution Consistency (*3)	96.6 ± 0.49
Entropy Preservation (*4)	96.2 ± 0.40
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	94.7 ± 1.49
Feature Accuracy (*10)	97.3 ± 0.64

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7525974025974025	0.7512987012987012	0.7538961038961038	0.0012987012987013102	
F1	0.6101915710164609	0.6135444370271858	0.6111216169297813	0.0009300459133203987	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	0.7272727272727273	0.7396103896103895	0.010389610389610393	
F1	0.592152063889281	0.595276772984698	0.6048526464664438	0.012700582577162823	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.7448051948051948	0.7577922077922078	-0.005844155844155841	
F1	0.6399852354635025	0.5948829827032875	0.6226383779186622	-0.017346857544840322	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7168831168831169	0.6987012987012986	0.7097402597402597	-0.007142857142857150 4	
F1	0.5757876188975466	0.570589240429687	0.5657696759007639	-0.010017942996782608	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6740259740259741	0.6454545454545454	0.6889610389610389	0.014935064935064923	
F1	0.4937730577787514	0.4677005369904908	0.45197134767571556	-0.041801710103035830	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	0.7558441558441558	0.7629870129870129	-0.00909090909090904	
F1	0.6262914445612621	0.6096525765106874	0.6149574605335918	-0.01133398402767033	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7480519480519481	0.740909090909091	0.7493506493506493	0.0012987012987013213	
F1	0.6188453461258308	0.6160799340121856	0.6246176798355273	0.005772333709696509	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.7402597402597403	0.7467532467532468	-0.007792207792207806	
F1	0.5998259463389416	0.5643608939509382	0.5939881206192099	-0.005837825719731682	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.7467532467532467	0.7616883116883116	0.005194805194805195	
F1	0.5733680734057942	0.5701674089377686	0.5932754999666259	0.01990742656083161	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7487012987012986	0.7616883116883116	0.012337662337662337	
F1	0.6076604413342895	0.6240759219173765	0.6368067453543482	0.02914630402005874	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	85.0	50.0	30.0	126.0	37.4	0.19	22.0	1.0	1.0
2	4.0	133.0	78.0	16.0	510.0	25.4	0.258	42.0	0.0	1.0
3	0.0	129.0	82.0	0.0	0.0	30.7	0.389	25.0	1.0	1.0
4	1.0	193.0	0.0	37.0	0.0	29.5	0.307	25.0	1.0	1.0
5	2.0	108.0	50.0	40.0	0.0	24.3	0.6	22.0	0.0	1.0
6	8.0	129.0	68.0	18.0	155.0	67.1	0.856	21.0	1.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	165.0	58.0	30.0	168.0	31.6	0.261	26.0	1.0	2.0
2	2.0	73.0	74.0	0.0	0.0	34.9	0.304	29.0	0.0	2.0
3	5.0	137.0	64.0	29.0	140.0	32.9	0.189	51.0	1.0	2.0
4	0.0	108.0	60.0	40.0	478.0	26.8	0.326	65.0	1.0	2.0
5	10.0	109.0	82.0	0.0	36.0	25.0	0.401	31.0	0.0	2.0
6	6.0	129.0	58.0	33.0	210.0	30.4	1.159	23.0	0.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	11.0	176.0	72.0	0.0	0.0	23.5	0.263	39.0	0.0	3.0
2	4.0	124.0	60.0	30.0	77.0	28.2	0.239	30.0	0.0	3.0
3	0.0	103.0	72.0	28.0	0.0	22.1	0.292	25.0	0.0	3.0
4	0.0	171.0	54.0	0.0	0.0	34.4	0.767	62.0	0.0	3.0
5	2.0	81.0	70.0	28.0	0.0	38.5	0.536	28.0	0.0	3.0
6	3.0	140.0	80.0	0.0	0.0	28.5	2.137	28.0	1.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	144.0	62.0	23.0	0.0	25.0	0.324	29.0	0.0	4.0
2	3.0	132.0	74.0	35.0	126.0	36.5	0.258	38.0	1.0	4.0
3	4.0	98.0	86.0	18.0	130.0	26.8	0.26	30.0	0.0	4.0
4	4.0	143.0	86.0	0.0	0.0	42.7	0.471	36.0	1.0	4.0
5	5.0	100.0	78.0	23.0	0.0	0.0	0.227	29.0	0.0	4.0
6	8.0	197.0	66.0	16.0	0.0	27.8	0.153	45.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	93.0	78.0	0.0	0.0	24.7	0.652	25.0	0.0	5.0
2	4.0	180.0	104.0	30.0	43.0	30.8	0.443	45.0	1.0	5.0
3	0.0	140.0	84.0	23.0	148.0	26.8	0.257	26.0	0.0	5.0
4	1.0	107.0	48.0	30.0	600.0	26.5	0.338	31.0	0.0	5.0
5	3.0	140.0	90.0	18.0	64.0	32.5	0.748	21.0	0.0	5.0
6	0.0	146.0	82.0	34.0	200.0	35.4	0.135	23.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	83.0	62.0	32.0	540.0	39.1	0.733	28.0	1.0	6.0
2	8.0	166.0	78.0	13.0	0.0	31.2	0.326	45.0	1.0	6.0
3	1.0	97.0	58.0	0.0	0.0	23.7	0.192	32.0	0.0	6.0
4	0.0	100.0	60.0	0.0	76.0	21.8	0.234	34.0	0.0	6.0
5	0.0	141.0	0.0	40.0	225.0	39.2	0.68	27.0	0.0	6.0
6	5.0	108.0	0.0	13.0	0.0	30.8	0.926	37.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	100.0	74.0	15.0	49.0	37.6	0.194	37.0	0.0	7.0
2	1.0	99.0	74.0	31.0	61.0	22.4	0.514	41.0	0.0	7.0
3	5.0	175.0	74.0	41.0	0.0	39.8	0.534	37.0	1.0	7.0
4	4.0	151.0	70.0	31.0	140.0	27.8	1.318	30.0	0.0	7.0
5	3.0	125.0	76.0	27.0	235.0	26.2	0.246	25.0	0.0	7.0
6	1.0	122.0	50.0	0.0	0.0	24.2	0.159	36.0	0.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	103.0	68.0	0.0	0.0	29.0	0.14	28.0	0.0	8.0
2	0.0	145.0	0.0	17.0	0.0	37.6	0.142	27.0	0.0	8.0
3	5.0	139.0	82.0	33.0	0.0	30.9	1.022	37.0	1.0	8.0
4	4.0	145.0	56.0	30.0	0.0	34.7	0.14	21.0	0.0	8.0
5	0.0	138.0	78.0	36.0	0.0	23.4	0.269	24.0	0.0	8.0
6	1.0	179.0	82.0	39.0	478.0	28.6	0.655	24.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	195.0	80.0	46.0	86.0	36.9	0.315	28.0	1.0	9.0
2	0.0	154.0	82.0	25.0	0.0	32.3	0.366	26.0	1.0	9.0
3	2.0	158.0	74.0	11.0	64.0	38.7	0.349	28.0	1.0	9.0
4	5.0	116.0	0.0	33.0	387.0	37.2	0.19	35.0	0.0	9.0
5	1.0	80.0	70.0	0.0	330.0	25.4	0.427	25.0	0.0	9.0
6	12.0	126.0	62.0	32.0	105.0	0.0	0.572	40.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	14.0	151.0	64.0	12.0	76.0	35.5	0.254	59.0	1.0	10.0
2	15.0	81.0	52.0	28.0	540.0	25.9	0.578	34.0	0.0	10.0
3	5.0	98.0	56.0	14.0	190.0	26.3	2.329	32.0	0.0	10.0
4	11.0	129.0	74.0	26.0	0.0	35.3	0.893	45.0	0.0	10.0
5	5.0	68.0	72.0	0.0	0.0	33.2	0.408	32.0	1.0	10.0
6	4.0	129.0	64.0	17.0	0.0	27.6	0.381	25.0	1.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan